Abhishek Kumar Singh

Embedded Systems | Edge AI | IoT
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PROFILE

ECE (IoT) student who built a C++ early-exit inference runtime with 25.2% lower mean latency in a controlled CPU benchmark. Hands-on ESP32-S3 signed telemetry and a reproducible controller-failure lab.

EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur

B.Tech, Electronics & Communication Engineering (IoT) | 2025-2029 (Expected)
First-year CGPA: **8.4/10**

Kendriya Vidyalaya, IIT Kanpur Campus | Class XII: **89%** | Class X: **89.4%**

SELECTED PROJECTS

AeroNode | Adaptive Edge-AI Inference Runtime

github.com/abhishekkr10/AeroNode

- Engineered staged C++17 inference with persistent ONNX Runtime sessions and confidence-based early exits, skipping unnecessary stages while checking predictions against reference execution.
- Measured **25.2% lower mean latency** (2.1052 ms vs 2.8139 ms) at 72.6857% vs 72.9400% accuracy on 1,966 held-out images (Windows CPU, batch 1, one thread, three repetitions).

CausalBench | Reproducible Controller-Failure Lab

github.com/abhishekkr10/CausalBench

- Created a local lab to reproduce sensor/controller failures using timestamped traces, seeded faults and baseline-versus-guarded decision replay with simulated actuation.
- Automated failure minimization, regression checks and source/evidence exports; documented validation with **93 Python tests and 13 browser workflows**.

SafeTrack | Signed IoT Shipment Monitoring | Team project

github.com/inamansingh2401/tamperproject (team repo)

- Designed the end-to-end architecture and wrote ESP32-S3 firmware for DHT11, MPU6050 and light sensing, with ATECC508A-signed, hash-linked events for tamper-evident shipment records.
- Built the Python gateway with durable SQLite outbox, FastAPI backend and PostgreSQL storage (migrated from Render to Supabase); contributed to Admin/User dashboard design.

JOCKY | Forensics Analysis Platform

github.com/abhishekkr10/jocky-forensics

- Built a Rust investigation compiler and evidence analysis tool with signed integrity checks; working local platform with a synthetic public demo.

TECHNICAL SKILLS

Embedded/Hardware: C, C++17, ESP32-S3, I2C, UART, MPU6050, DHT11, ATECC508A

ML/Inference: Python, PyTorch, ONNX, ONNX Runtime

Backend/Tools: FastAPI, PostgreSQL, SQLite, ZeroMQ, React, TypeScript/JavaScript, Rust, CMake, Git/GitHub